

"Adversarial Hubness in Multi-Modal Retrieval" Analysis for IEEE S&P'26

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1. Background — Multi-Modal Retrieval & Hubness

Multi-Modal Retrieval

- aligns different modalities (image, text, audio, etc.) into a single embedding space, and retrieves items based on **cosine similarity** (ImageBind(CVPR 2023), CLIP(ICML 2021), etc.)

Hubness

- due to the **curse of dimensionality** in high-dim embedding spaces, a small number of points **repeatedly appear as the nearest neighbor** of many **semantically unrelated queries** [56]

Natural hub

- occurs by chance, only slightly more often: the top-1 result for only 102 of 25,000 queries (about 0.4%)

Can hubness be created **"intentionally"** rather than **"by chance"**?

→ **Adversarial hub**

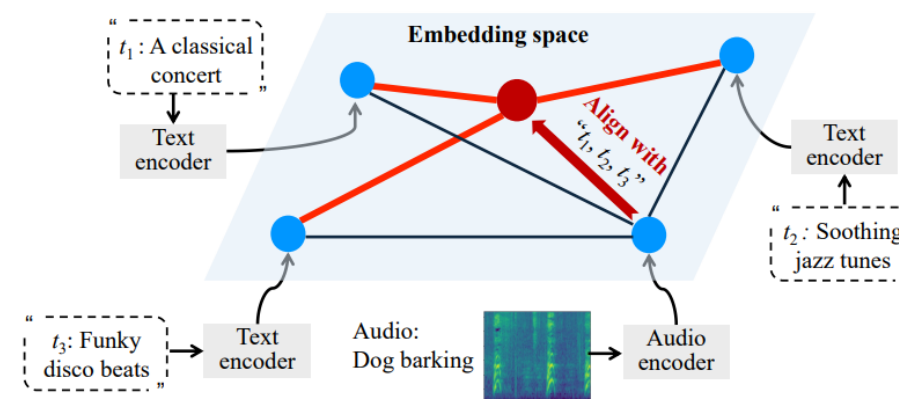


Figure 1: A cross-modal adversarial hub.

2. Attack Structure — Adversarial Hub

Structure of an Adversarial Hub:

g_c (**carrier**): plausible-looking content chosen by the attacker (ex. image)

δ (**perturbation**): imperceptible noise added by the attacker. $\|\delta\|_\infty \leq \epsilon$ (use ℓ_∞ bounds)

$g_a = g_c + \delta$: the resulting adversarial hub

Optimized to **have high similarity with many semantically unrelated queries** at once

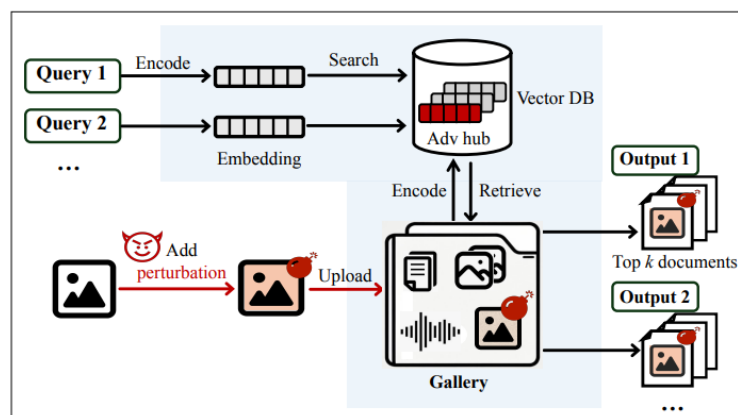


Figure 5: **Attacking retrieval system with an adversarial hub.**

1. g_a is injected into the gallery (Vector DB)
2. for diverse queries, g_a is repeatedly retrieved as top-1 resulting in spreading spam/misinformation or polluting a specific topic
→ retrieved as top-1 for **21,000 out of 25,000 queries**
(the most common natural hub is 102 out of 25,000 queries (about 0.4%))

3. Threat Model

Attacker Knowledge (model access)

- **White-box**: full access to model architecture and parameters
- **Black-box, Transfer attack**: craft on a **surrogate model**, hoping it transfers to the target
- **Black-box, Query attack**: **no gradient** access, only observes the target API's similarity scores (score-based)

Attacker's Goal

- **Indiscriminate (Universal)**: a hub that hits **as many queries as possible**
- **Concept-specific**: a targeted hub that only hits **queries related to a specific topic**
 - **Word-based**: its target query set is defined **lexically** — queries that contain a specific word
 - **Cluster-based**: target query set is defined via **embedding similarity** — already grouped in embedding space

Knowledge of query distribution

- **Exact**: access to real target queries, a strong assumption
- **Proxy**: a similar public dataset, a weaker and more realistic assumption

→ Authors Experimented on almost all combinations

4. Generating Adversarial Hubs — Attack Method

Target compression

- the entire query set Q_t is collapsed into a single **centroid** c_t (Arithmetic mean), simplifying the optimization target and enabling scalability

Optimization: PGD (Projected Gradient Descent, based on Madry et al., ICLR 2018)

$$c_t = \frac{1}{|Q_t|} \cdot \sum_{q \in Q_t} \theta(q)$$

$$L(\delta) = -\cos(\theta(g_c + \delta), c_t)$$

$$\delta^{(t+1)} = \Pi_{\|\delta\|_\infty \leq \epsilon} \left[\delta^{(t)} - \alpha \cdot \text{sign} \left(\nabla_\delta L(\delta^{(t)}) \right) \right]$$

- Moves by a fixed step **using only the sign of the gradient**
- **White-box** uses direct **gradients with the accurate model**
- **Transfer attack** optimizes the average **loss across multiple surrogate models**
- **Query attack**: adapts **Square Attack** to improve similarity gradually via **gradient-free random search** (up to 100,000 q)
 - **Square Attack (Andriushchenko et al., ECCV 2020)**
 - : each iteration, applies a $\pm \epsilon$ **perturbation to a random square patch** and keeps it only if similarity improves. originally to induce misclassification, and repurposed here by **swapping the objective function to similarity maximization**

5. Concept-Specific vs Universal Attack

	Universal (Indiscriminate)	Concept-Specific (Word-based)	Concept-Specific (Cluster-based)
Target definition	100 random queries (entire set)	Queries containing a specific word	Embedding clusters via K-means
Goal	Dominate as many queries as possible	Targeted at a specific topic's queries only	Targeted at a specific topic's queries only
Attack success rate	84%+ (vs. 0.4% for natural hub)	28.6-61.7%	About 100%
Collateral Damage (false hits on unrelated queries)	Easy to detect (spreads broadly)	0.7% (very low)	18.4%

TABLE 4: **Concept-specific adversarial hubs on MS COCO with and without QB-Norm defense.** Performance is reported on poisoned galleries using *Original* and *Targeted* query sets. Standard deviations omitted for concept-specific hubs due to evaluation on heterogeneous query subsets.

QB-Norm	Retrieved	Query	$R@1$ (%)	$R@5$ (%)	$R@10$ (%)	MdR
No	Relevant Doc.	Original	10.9 ± 2.0	69.6 ± 0.0	79.8 ± 0.0	2.9 ± 0.1
	Universal Hub	Original	85.1 ± 3.1	98.1 ± 0.7	99.0 ± 0.4	1.0 ± 0.0
	Word-based Concept-specific Hub	Original	0.2	0.4	0.7	912.1
	Word-based Concept-specific Hub	Targeted	28.6	53.5	61.7	50.8
	Cluster-based Concept-specific Hub	Original	8.6	15.0	18.4	24.1
	Cluster-based Concept-specific Hub	Targeted	100.0	100.0	100.0	1.0
Yes	Relevant Doc.	Original	49.7 ± 0.0	74.1 ± 0.1	82.6 ± 0.1	2.0 ± 0.0
	Universal Hub	Original	0.0 ± 0.0	11.5 ± 2.4	12.2 ± 2.7	699.9 ± 72.8
	Word-based Concept-specific Hub	Original	0.0	0.2	0.3	1379.2
	Word-based Concept-specific Hub	Targeted	22.9	44.6	53.6	55.5
	Cluster-based Concept-specific Hub	Original	0.1	5.1	7.6	4015.7
	Cluster-based Concept-specific Hub	Targeted	95.3	98.3	99.0	1.0

trade-off: **the stronger the attack(Universal), the less stealthy it becomes**

- In practice a stealthy concept-specific attack that mostly evades defenses is the more rational choice over an easily-detected universal attack

6. Experiments and Evaluation

Models: ImageBind (6 modalities) (CVPR 2023), AudioCLIP (3 modalities) (ICASSP 2022), CLIP (2 modalities) (ICML 2021)

Datasets: MS COCO (text-image), CUB-200-2011 (image-image), AudioCaps (text-audio), MSR-VTT (video-text, proxy)

TABLE 1: **Result for text-to-image retrieval (MS COCO).** “Clean” denotes the original gallery, “Poisoned” denotes the gallery with an adversarial hub. “Relevant Doc.” is the documents relevant to the query, “Adv. Hub.” is the adversarial hub. Attack performance is shaded. Results show mean and standard deviation across 100 independently generated adversarial hubs. See Section 4 for detailed description of metrics and evaluation methodology.

Model	Gallery	Retrieved	$R@1$ (%) $\uparrow$	$R@5$ (%) $\uparrow$	$R@10$ (%) $\uparrow$	MdR $\downarrow$
ImageBind	Clean	Relevant Doc.	48.4	72.7	81.0	2.0
	Poisoned	Relevant Doc.	10.9 $\pm$ 2.0	69.6 $\pm$ 0.0	79.8 $\pm$ 0.0	2.9 $\pm$ 0.1
	Poisoned	Adv. Hub	85.1 $\pm$ 3.1	98.1 $\pm$ 0.7	99.0 $\pm$ 0.4	1.0 $\pm$ 0.0
AudioCLIP	Clean	Relevant Doc.	18.8	39.5	50.7	10.0
	Poisoned	Relevant Doc.	8.8 $\pm$ 3.5	36.9 $\pm$ 0.4	49.3 $\pm$ 0.2	11.0 $\pm$ 0.0
	Poisoned	Adv. Hub	62.0 $\pm$ 18.7	81.4 $\pm$ 13.3	87.5 $\pm$ 10.4	1.0 $\pm$ 1.4
OpenCLIP-ViT	Clean	Relevant Doc.	48.5	72.8	81.1	2.0
	Poisoned	Relevant Doc.	11.3 $\pm$ 2.2	69.8 $\pm$ 0.0	79.9 $\pm$ 0.0	3.0 $\pm$ 0.1
	Poisoned	Adv. Hub	84.4 $\pm$ 3.5	97.9 $\pm$ 0.8	98.9 $\pm$ 0.5	1.0 $\pm$ 1.0
OpenCLIP-RN50	Clean	Relevant Doc.	28.3	53.0	64.1	5.0
	Poisoned	Relevant Doc.	1.4 $\pm$ 0.8	49.8 $\pm$ 0.0	62.2 $\pm$ 0.0	6.0 $\pm$ 0.0
	Poisoned	Adv. Hub	97.6 $\pm$ 1.6	99.5 $\pm$ 0.5	99.7 $\pm$ 0.3	1.0 $\pm$ 0.0

TABLE 2: **Text-to-audio retrieval (AudioCaps).**

Model	Gallery	Retrieved	$R@1$ (%) $\uparrow$	$R@5$ (%) $\uparrow$	$R@10$ (%) $\uparrow$	MdR $\downarrow$
ImageBind	Clean	Relevant Doc.	10.6 $\pm$ 0.0	30.7 $\pm$ 0.0	43.7 $\pm$ 0.0	13.0 $\pm$ 0.0
	Poisoned	Relevant Doc.	7.3 $\pm$ 0.6	27.4 $\pm$ 0.3	40.5 $\pm$ 0.1	16.0 $\pm$ 0.0
	Poisoned	Adv. Hub	52.2 $\pm$ 6.7	71.2 $\pm$ 5.6	81.3 $\pm$ 4.8	1.5 $\pm$ 0.7
AudioCLIP	Clean	Relevant Doc.	6.2 $\pm$ 0.0	20.7 $\pm$ 0.0	31.8 $\pm$ 0.0	27.0 $\pm$ 0.0
	Poisoned	Relevant Doc.	2.5 $\pm$ 0.8	18.8 $\pm$ 0.3	30.6 $\pm$ 0.2	28.0 $\pm$ 0.0
	Poisoned	Adv. Hub	58.0 $\pm$ 13.7	77.6 $\pm$ 10.5	85.8 $\pm$ 7.9	1.5 $\pm$ 0.9

TABLE 3: **Image-to-image retrieval (CUB-200-2011).**

Model	Gallery	Retrieved	$R@1$ (%) $\uparrow$	$R@3$ (%) $\uparrow$	$R@5$ (%) $\uparrow$	$R@10$ (%) $\uparrow$	MdR $\downarrow$
ImageBind	Clean	Relevant Doc.	59.2	77.7	84.9	92.3	1.0
	Poisoned	Relevant Doc.	55.7 $\pm$ 0.2	75.5 $\pm$ 0.1	83.2 $\pm$ 0.1	91.0 $\pm$ 0.0	1.0 $\pm$ 0.0
	Poisoned	Adv. Hub	5.5 $\pm$ 0.7	39.5 $\pm$ 2.6	62.2 $\pm$ 2.5	87.7 $\pm$ 2.0	4.2 $\pm$ 0.4
AudioCLIP	Clean	Relevant Doc.	12.1	22.2	28.3	38.9	19.0
	Poisoned	Relevant Doc.	11.0 $\pm$ 0.5	21.6 $\pm$ 0.5	27.9 $\pm$ 0.4	39.2 $\pm$ 0.2	18.0 $\pm$ 0.0
	Poisoned	Adv. Hub	13.3 $\pm$ 6.1	31.6 $\pm$ 11.1	44.8 $\pm$ 12.9	66.7 $\pm$ 13.1	6.8 $\pm$ 3.1
OpenCLIP-ViT	Clean	Relevant Doc.	59.5	79.5	86.2	93.0	1.0
	Poisoned	Relevant Doc.	56.6 $\pm$ 0.2	77.2 $\pm$ 0.2	84.4 $\pm$ 0.1	92.6 $\pm$ 0.1	1.0 $\pm$ 0.0
	Poisoned	Adv. Hub	4.1 $\pm$ 0.6	34.6 $\pm$ 2.7	61.0 $\pm$ 3.1	90.7 $\pm$ 2.6	4.7 $\pm$ 0.5
OpenCLIP-RN50	Clean	Relevant Doc.	22.0	38.7	48.0	62.1	6.0
	Poisoned	Relevant Doc.	20.5 $\pm$ 0.4	36.6 $\pm$ 0.3	46.4 $\pm$ 0.1	60.8 $\pm$ 0.1	6.6 $\pm$ 0.5
	Poisoned	Adv. Hub	20.0 $\pm$ 2.8	45.7 $\pm$ 4.0	59.4 $\pm$ 3.7	77.7 $\pm$ 2.9	4.1 $\pm$ 0.6

TABLE 6: **Pinecone image-to-image retrieval (CUB-200-2011).**

Gallery	Retrieved	$R@1$ (%) $\uparrow$	$R@3$ (%) $\uparrow$	$R@5$ (%) $\uparrow$
Clean	Relevant Doc.	25.4	43.0	52.7
Poisoned	Relevant Doc.	24.5 $\pm$ 0.2	41.9 $\pm$ 0.6	51.8 $\pm$ 0.7
Poisoned	Adv. Hub	7.5 $\pm$ 1.5	21.4 $\pm$ 3.5	31.7 $\pm$ 4.5

– Cross-modal attacks (text query-image hub) are more effective than uni-modal ones (image query-image hub)

- Baseline similarity across different modalities is originally very low (about 0.014), so even a small increase produces a relatively dramatic jump
- Within the same modality, the baseline is already high (around 0.2-0.3), so competition is fierce and it is harder for a hub to stand out

6. Experiments and Evaluation

Ablation Studies

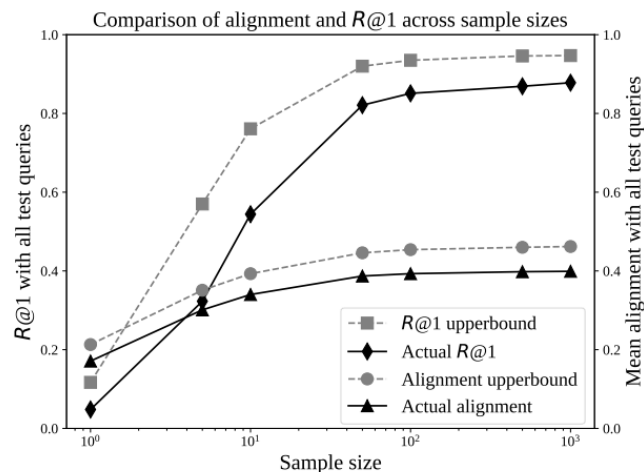


Figure 8: **Performance vs. sample size.** Attack success rate (ASR) and cosine similarity between adversarial hub and query embeddings improve with larger target sample sizes, converging at around 100 samples.

TABLE 8: **Impact of perturbation bound ϵ on attack performance.** Results show how different perturbation magnitudes affect the retrieval metrics for adversarial hubs generated with ImageBind on MS COCO dataset.

ϵ	$R@1(\%) \uparrow$	$R@5(\%) \uparrow$	$R@10(\%) \uparrow$	$MdR \downarrow$
8/255	62.8 $\pm$ 10.7	92.9 $\pm$ 4.4	96.4 $\pm$ 2.4	1.1 $\pm$ 0.4
16/255	85.1 $\pm$ 3.1	98.1 $\pm$ 0.7	99.0 $\pm$ 0.4	1.0 $\pm$ 0.0
32/255	90.5 $\pm$ 1.4	98.8 $\pm$ 0.3	99.4 $\pm$ 0.2	1.0 $\pm$ 0.0
64/255	91.5 $\pm$ 1.1	99.0 $\pm$ 0.3	99.5 $\pm$ 0.2	1.0 $\pm$ 0.0

TABLE 9: **Impact of different geometric centers on attack performance.** Results show how different choices of geometric center affect the retrieval metrics for adversarial hubs generated with ImageBind on the MS COCO dataset.

Center	$R@1(\%) \uparrow$	$R@5(\%) \uparrow$	$R@10(\%) \uparrow$	$MdR \downarrow$
Centroid	85.1 $\pm$ 3.1	98.1 $\pm$ 0.7	99.0 $\pm$ 0.4	1.0 $\pm$ 0.0
Trimmed centroid	84.0 $\pm$ 3.6	97.8 $\pm$ 0.8	98.8 $\pm$ 0.5	1.0 $\pm$ 0.0
Medoid	11.8 $\pm$ 2.1	22.7 $\pm$ 5.2	29.2 $\pm$ 6.6	44.2 $\pm$ 16.7
Geometric median	85.3 $\pm$ 3.5	98.1 $\pm$ 0.6	99.0 $\pm$ 0.4	1.0 $\pm$ 0.0

perturbation $\epsilon=16/255$ (or 0.05 in audio), sample size $|Q_t|=100$ is reasonable choice

7. Existing Defense Mechanisms

QB-Norm(CVPR 2022) / NNN(EMNLP 2024) - Natural Hubness Mitigation Method

- flags gallery items that repeatedly appear in the top-k across many query-bank queries as **suspected hubs**, then **renormalizes similarity**
 - Universal hubs:** substantially **mitigated by diligent QB-Norm**, but defense effectiveness drops sharply with the low-cost approximate version (NNN-style)
 - Concept-specific hubs:** effectively **defeat QB-Norm entirely**, since they are unrelated to the majority of generic query-bank queries and **only respond to a narrow concept, so renormalized slightly**.

Adversarial-example-style defenses (JPEG compression, feature squeezing, feature distillation)

- Cannot selectively remove just the hub's perturbation. they **indiscriminately degrade every input**
- **Cannot defend an adaptive attack** that simulates the defense as an augmentation step during hub generation, R@1 remains 24.6%, still effective

Adversarial training / Certified methods

- Computationally infeasible at the scale of CLIP/ImageBind, largely limited to small CIFAR-10-scale models

8. Challenges

Simultaneously capturing many different queries in high dimensions

- Semantically unrelated queries are naturally far apart in the embedding space, so finding a single center point close to all of them at once within the epsilon-ball constraint is a geometrically hard problem

Robustness-Invariance Trade-off

- For retrieval to work well, the model must be sensitive to small differences in input, and that very sensitivity is the root reason an attacker can push the embedding in a desired direction using gradients

Modality-agnostic reach

- optimize only images/audio to attack both unimodal and multimodal (cross-modal) queries
- 다만 이산적인 text와 달리 연속적인 이미지와 오디오는 부자연스럽지 않게 perturbate할 수 있어 stealthy할 수 있다.

9. Contributions

First systematic demonstration that natural hubs can be created intentionally

- Generalizability: shows that natural hubs can be converted into adversarial hubs

Universal adversarial hub

- retrieved by over 100x more queries than a natural hub (21,000 vs. 102 of 25000 queries in MS-COCO)

Concept-specific attacks

- achieves precise targeting and stealthiness at the same time
- existing natural-hubness mitigation techniques (QB-Norm/NNN) are ineffective against concept-specific hubs

Task-agnostic attack

- targets the embedding itself rather than a specific downstream task (classification, jailbreaking, etc.)
- so it can propagate to any downstream task that relies on the embedding, including retrieval, classification, and generation

10. Limitations

Only covers retrieval poisoning

- only addresses making the hub get retrieved, and **does not address generation poisoning**, the second stage of RAG poisoning where the retrieved hub causes a generative model to produce harmful output

Limits of black-box performance

- **less effective than white-box or the best transfer attacks**. It works generically without model-architecture knowledge, but is not optimal

Ablation Studies experimented on only MS-COCO + ImageBind embedding

Transferability depends on architectural similarity

- transfers well between encoders of the same family (ViT-based), but transfers **poorly to heterogeneous architectures** such as ResNet or separate audio branches

11. Motivation & Future Research Directions

- **Background:** having no prior hands-on research experience in AI, I analyzed this paper while exploring *Breaking AI's Protective Layers (Alignment, Watermark, Blur, etc.)* as a candidate research topic
- **I enjoyed reading this paper, and organized the following personal thoughts**
 - Would like to implement an adaptive attack myself, building defense-awareness directly into the hub-generation process so it bypasses known defenses
 - **Idea for comparing embeddings across open-source multi-modal embedding models:** if embeddings turn out similar across models, transfer attacks should work well; if they differ, I'm curious why different models perceive the same word or image so differently from humans. Though if they differ too much, I also wonder whether the current surrogate-based attack approach would even work at all
 - Confirmed that model-fingerprinting research for CLIP-like models is relatively lacking compared to LLMs, highlighting the difficulty of estimating a target's architecture
- **This direction, attacking AI embeddings and representations and studying transferability, is one I'm continuing to consider for future research topics**

12. More Interested Papers (in IEEE S&P'26, USENIX'26)

#	Category	Title	Conference	Content
1	Breaking AI's Protective Layers	Adversarial Hubness in Multi-Modal Retrieval	IEEE S&P'26	Exploits embedding hubness to inject content into unrelated queries (today's paper)
2	Tracing the Attack Back to Its Source	Who Taught the Lie? Responsibility Attribution for Poisoned Knowledge in RAG	IEEE S&P'26	Traces poisoned texts responsible for RAG misgeneration
3	Exploiting the AI Supply Chain	Parasites in the Toolchain: A Large-Scale Analysis of Attacks on the MCP Ecosystem	IEEE S&P'26	Privacy-leakage attack propagating through the MCP toolchain without direct interaction with victim
4	Exploiting LLM Internal Mechanisms	One Bad Token Spoils the Barrel: Assessment, Detection, and Remediation of Glitch Tokens in LLMs	USENIX'26	Detects/remediates glitch tokens that bypass safety mechanisms
5	LLM Unlearning Robustness	LLM Unlearning Should Be Form-Independent	IEEE S&P'26	Identifies format-dependence in LLM unlearning, proposes fix

The 5 papers above are the ones I personally numbered (1-5) in my "Interested AI Security Paper List", planning to go through them one category at a time

Thank you

Q&A